

Econometrics III
Lindo, Fall 2013
Assignment 5
Due: xxxxx

This is intended to be competed on your own. If you are found to have submitted any code that is not your own, your continuation in the class will be subject to the outcome of a formal student conduct review.

This exercise has you consider the importance of the common trends (in outcomes) assumption that underlies difference-in-differences estimates. In particular, you will estimate the following model:

$$Y_{it} = \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 Treat_i Post_t + \epsilon_{it} \quad (1)$$

where T_i is an indicator variable taking the value 1 if individual i is in the treatment group and zero otherwise, $Post_t$ is an indicator variable taking the value 1 if the time period is from the post-treatment period, and $\epsilon \sim N(0, 6)$. The parameter of interest, of course, is β_3 which describes how outcomes in the post period for the treatment group diverge from what is expected based on their outcomes in pre period and based on the passage of time.

Consider a panel of 400 individuals (i.e., $i = 1, \dots, 400$) over ten periods (i.e., $t = 1, \dots, 10$) where half of the individuals are treated (i.e., $T_i = 1$), treatment occurs between periods 5 and 6, $\beta_0 = 1, \beta_1 = 0.5, \beta_2 = 0.5$, and the true treatment effect, $\beta_3 = 1$.

In order to examine how the estimate will be biased when the treatment group and control group have different trends, introduce a trend to the treatment group over -.6(.2).6, running 500 replications for each. (To be clear, this implies that you will be running 7 simulations involving 500 replications.) Summarize these results by producing a table that includes the mean value of β_3 across each trend considered in addition to the proportion of times one would reject the null hypothesis $H_0 : \beta_3 = 1$ at the five-percent level.

Submit the do-file to your TA electronically. If it does not run on our computers unaltered, it will be returned to you for revision. In order to facilitate this, you may want to include the following at the top of your code:

```
clear
capture cd "<insert what you want to use as your working directory for this assignment here>"
if c(username)=="jlindo" {
    cd "~/Dropbox/Teaching/Microeconometrics (Grad)/Assignments/Assignment 5 DiffInDiff Trends"
}
if c(username)=="apackham" {
    cd "C:\Documents and Settings\apackham\Desktop\ECMT677/Assignments/Assignment 5 DiffInDiff Trends"
}
```